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Unit- 04 (COA)

1. Basic Concept Of Memory [AKTU-2022-23]
2. 2D and 2.5D Memory [AKTU-2020-21]
3. ROM memories
4. Cache memories
5. Auxilliary memories [AKTU-2022-23]
6. Magnetic disk and Magnetic Tape
7. Virtual memory [AKTU-2021-22, 2022-23]
8. Semiconductor memories
9. Associative Memory [AKTU-2021-22

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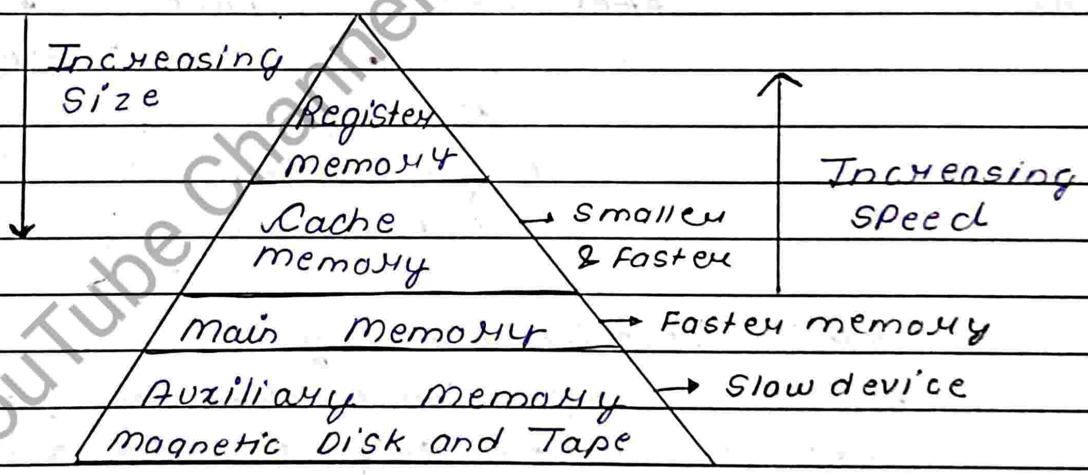
Memory :-

- Any Physical device used to store data, information or instruction temporily or Permanently
- Memory block is Split into a small no of Component known as Cells
- Each Cells have a Unique address to Store data in memory ranging From zero to memory Size-1

Memory Hierarchy



Main Memory

→ Central Storage Unit is Computer System

→ Fast Memory used to store Program & data during execution

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Semiconductor Memories :-

→ For Storing digital data.

→ Small Size, Large Capacity, reliable Working

Semiconductor memories :
- RAM
  - Static RAM
  - Dynamic RAM
- ROM
  - Masked Programmable ROM
  - PROM
  - EPROMs
  - EAPROMS

1. RAM (Volatile Memory)
Memories that store data temporarily, data is Stored only till the duration Power Supply to the IC is ON. Once the Supply gets OFF then the Stored data get lost

2. ROM (Non-Volatile Memory)
Data retained in memory even if the Power Supply is OFF ∴ the data Store Permanently

- Static RAM
Array of flip-flops that are used to Store data. Hold data until the Power Supply is ON. Faster

- Dynamic RAM
It is also a read/write memory that Stores data in form of Capacitor and transistor pairs. Stored Charge take some millisecond to get dissipated so periodic refressing of element is required

- Masked Programmable ROM
It is also known as ROMs. In this data is Store Permanent (This is inbuild Feauture given by the manufucture).

- PROM (Programmable Read Only Memory
PROM is a Programmed memory. Programmed by the user but these Offers re-Programmability after erasing loaded data Content

- EPROM (Erasable Programmable ROM)
It Can be Programmed number of times. In this we use UV light to erase the data
Memory erasing time lies b/w 10 to 30 minutes

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⊙ EAPROMS (Electrically Altered Programmable Read Only Memory)

→ In this we apply electrically signal to erase the data.

→ Initially data is erased by apply external voltage at the erase pin of chip. So it somewhat increase the complexity of overall system

2D RAM	2.5D RAM
• In 2D RAM organ^n a RAM of $2^n \times m$ consisting 2^n memory words of m-bits.	• In 2.5D RAM organ^n the no of addr^n lines are divided into approximately equal parts, one for row select decoder and another for column
• It has a row select decoder of size $n \times 2^n$ to select out of 2^n memory words	• In 2.5D RAM, hardware is variable
• Each row have m column	• It required less no of logic gates
• It required more no of logic gate	

• 2D RAM is more Complex	• 2.5D RAM is less Complex

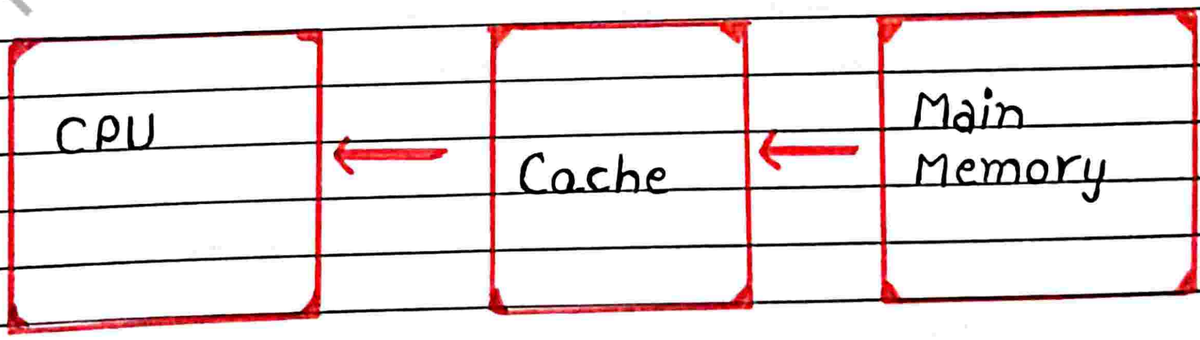
Cache Memory :-

- Cache Memory is a Special Very High-Speed memory
- It is Costlier than main memory but economical than CPU register
- Cache memory is an extremely Fast memory type that acts as a buffer between RAM and the CPU

Why Cache is Needed :-

- The Cache memory is required to balance the Speed mismatch between the main memory and the CPU

- The clock of the Processor is Very Fast while the main memory access time is Comparatively slower



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Cache Memory Performance Issue :-

- If the CPU needs to read data / Instⁿ From memory, the First Cache is examined. If the word is Found in Cache, it is known as Cache Hit

- The Performance Of Cache memory is Frequently measured in term of a quality Called Hit Ratio.

Cache Memory Mapping

1. Associative Mapping :-

→ Fastest and most Flexiable Cache Organization uses Associative memory.

→ In Associative mapping, caches are made up of Associative memory. Associative memory is used to store both the address and Context (data) of the memory word.

→ It Permit any location in Cache to Store any word From main memory

2. Direct Mapping :-

→ Associative memories are expensive Compared to RAM.

→ For RAM Direct Mapping can be Used.

→ It is a Simplest technique

Note: In main memory is of 2^n words and Cache memory 2^k words.

3. Set Associative Mapping :-

→ Improved From the direct mapping, where drawback of direct mapping is removed.

→ Two words with the Same Index in address but with different tag Cannot be Stored in Cache memory at Same time (Drawback)

→ In set Associative mapping each word of a Cache can Store two or more words of memory under the index address, Creating a set

→ Each data word is Stored together with its tag

Word-length = 2 (Tag + Data)

Virtual-Memory :-

→ It appears to be Present but actually it is not. It Provide illusion of a Large memory

→ Virtual Memory is the Concept that gives the illusion to the user the that they will have main memory is equal to the Capacity of Secondary Storage media.

Need of Virtual-memory

→ Virtual memory is a imaginary memory we are assuming. If we have a Program that exceed the size of main-memory than we need to use the Concept of Virtual Memory

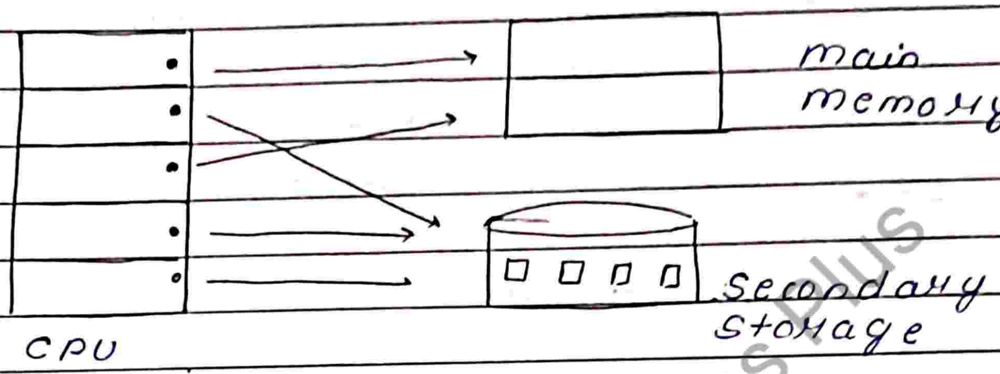
→ It is a temporary memory which is used along with the RAM of the System. It used Swapping

The size of Virtual memory

→ The Amount of Secondary memory Available

→ The addressing Scheme of the

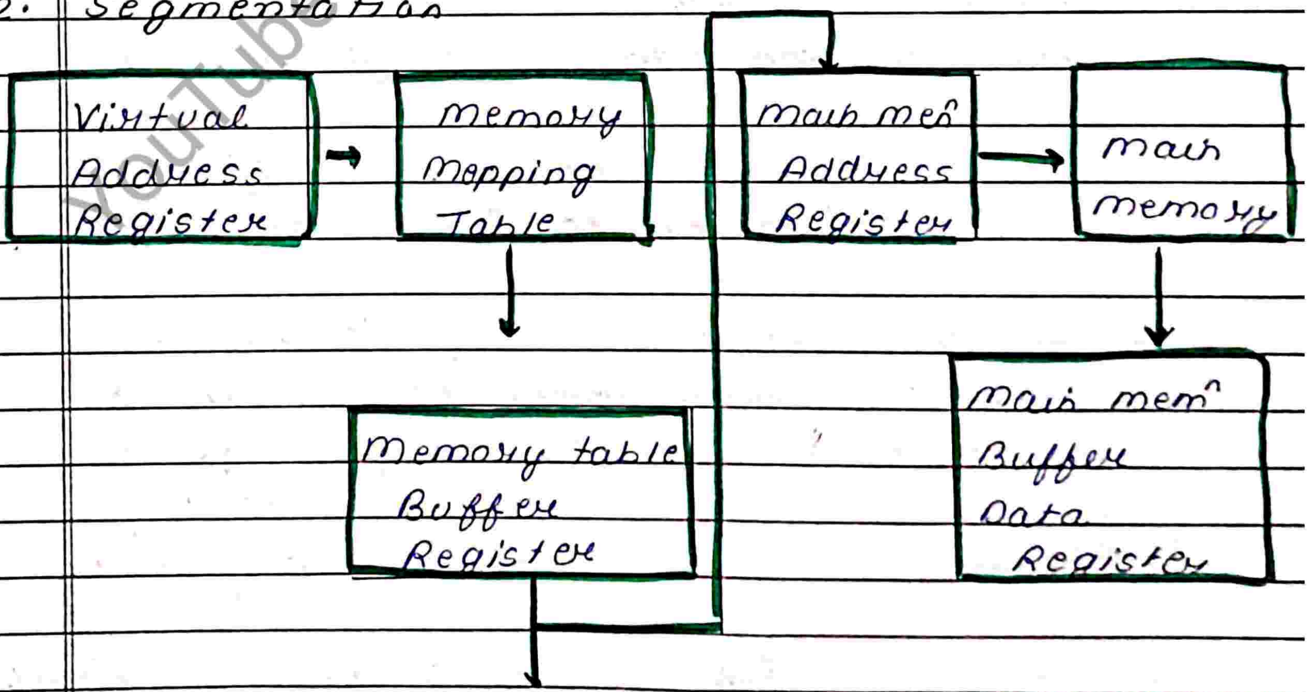
Computer System.



→ Virtual address is the address used by the Programmer & the set of such address is called address space or Virtual memory. An address in main memory is called Physical address.

Virtual memory Techniques

1. Paging
2. Segmentation



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→ In Paging address Space and memory Space is divided into equal Size gp
- Address Space (Pages)
- memory Space (Block)

Page Size = Block Size

Associative Memory

→ A memory unit acessed by Content is Called an Associative memory

→ Associative memory is accessed Simult-aneously & in Parallel on the basis of data Content rather by Specific address or Location

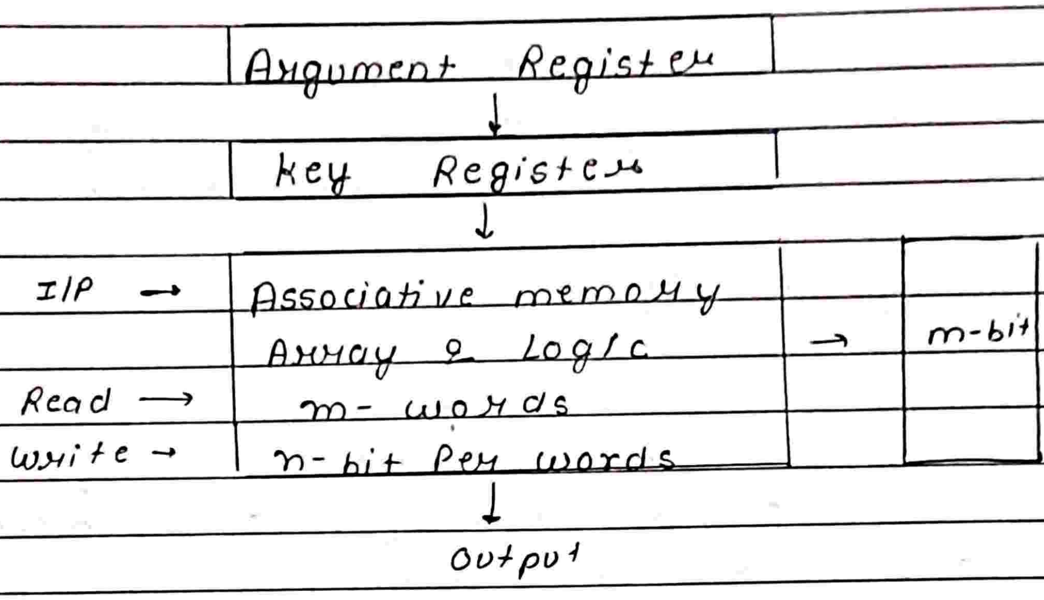
→ When a word is written in an associⁿ memory no address is given

→ The memory is Capable of finding an empty location to Store the word

→ An Associative memory is more expensive than RAM becoz each cell must have Storage Capability as well as logic ckt for matching Context with an external argument

* It is Useful where Search time is Critical

PSEUDONYMOUS



Auxilliary memory :-

- Auxilliary memory holds data For Future Use and retains inform^n even the Power Fails. Program and inform^m are Preserved For long-term Storage. or
- It is Lowest Cost, highest Space and Slowest approach storage in Computer System.

Magnetic Disk :-

- Magnetic Disk Provide a large amount of Secondary Storage to modern Computer
- It Stores data in the Form of tracks. SPorts and Sector

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→ A set of same tracks on all disks forms a cylinder. Each disk has its own read/write head

Magnetic Tape :-

→ They are generally used to store back-up data that is not frequently used or to transfer data from one system to another

→ Due to their sequential nature, magnetic tapes are not suitable for data files that need to be revised or updated frequently.